PimpriChinchwad Education Trust's
PimpriChinchwad College of Engineering
Sector No. 26, Pradhikaran,
Nigdi, Pune – 411 044

Dept: AS&H A.Y. 2025-26 Sem: I

Subject**: Statistical Data Analysis Using R (BSH23OE04)**

Class: SY B Tech(Comp. , IT, CSE-AIML and Regional)

Unit I: Understanding the data and Installation of R

Sr. No.	Question	Difficulty level
1.	What is data? What is the difference between data and information? Explain importance of data analysis.	Moderate
2.	What is data analysis? Explain steps of data analysis.	Moderate
3.	What are the methods of data collection? Describe in details.	Moderate
4.	What is the difference between quantitative and qualitative data? Explain their subtypes.	Moderate
5.	What are advantages of R programming amongst all.	Moderate
6.	What are applications of Data Analysis? Explain in details.	Moderate
7.	Give introduction of R software and describe GUI of R-studio	Moderate
8.	What is difference between R software and R-Studio	Moderate
9.	Explain all four windows of R-studio along with their functioning	Moderate
10.	Write in detail about 'Environment Window'. In a dataset what is the meaning of observation and variable.	Moderate
11.	Write the step wise process of primary data collection for any one of the following topic a. Effect of Social media on young generation b. Health habits of working women	DIFFICULT
	Coding based questions	
12.	Create a data frame as below **Name .student** | **Percentage** | **Class** Namita | 65 | First Aaroh | 90 | Distinction Rohit | 46 | Pass	Moderate

13.	How can you load a .csv file in R? Write syntax of the same.	Easy
14.	Write syntax to check inbuilt dataset in R; also write the command to check details of any dataset out of it.	Easy
15.	How do you install and load a package in R?	Easy
16.	How can you load a .xls/.xlsx file in R?**OR** How to import an Excel or CSV file in R, write all the steps in detail?	Easy
17.	Write syntax to check inbuilt dataset in R ;also how to view a particular dataset	Easy
18.	Tell the use of any two of folEasying commands from base R package a. str(dataset name) b. summary(dataset name) c. View(dataset name)	Easy
19.	data.frame() is important command in base R package. Explain this command with syntax and one example	Easy

20.	Create vectors as "Prn No.", "Name", "Bank Account", "Fees paid", "Mobile No." a data frame in R for the folEasying table	Moderate

Prn No.	Name	Bank Account	Fees paid	Mobile No.
123B1B01	Asha	IOB2205	147503	98000046
123B1C55	Kiran	IDBI3001	22572	54002456
123B1D12	Rajesh	SBI4451	3759	78940069
123B1D87	Salman	AXIS002	152007	55987800
123B1L06	Juhi	IOB2503	3759	22564200
123B1P65	Ragini	ICICI123	154053	23569800

21.	Write syntax to create a data frame containing 4 column that includes Roll number 1-10 , Marks of 3 subjects Math, English, Hindi & Marathi (* assume marks)	Moderate
22.	Write the script/commands for the following ● How to access inbuilt dataset in R? ● Select one of the dataset 'storms' and View it ● Which commands can be used to find structure of the dataset 'storms' and also write the command to find the statistical summary of the same dataset. ● Write the command to get first few observations of 'storms' data set	Difficult
23.	Create a script (syntax) for the following steps: I. Import a csv file named 'car_price_prediction' saved in D drive in folder SDAUR and check structure of it. II. Check data type of all variables. III. Change 'Levy'variable from character to double IV. Check first 10 entries of the data. V. Download the file to desktop.	DIFFICULT

<table>
<tr><td>24.</td><td>Access the inbuilt data ‘airquality’ and answer the following questions:
Whether this data set has NA values?
How many NA values are there? At what places?</td><td>DIFFICULT</td></tr>
<tr><td>25.</td><td>Access the inbuilt data ‘airquality’ and answer the following question:
What is minimum, maximum and average values of ‘Ozone’ and ‘Solar.R’ variables</td><td>Moderate</td></tr>
<tr><td>26.</td><td>Explain working of the following commands with syntax
1. rbind 2. as.factor 3. Replace_na 4.class 5. is.numeric 6.cbind()</td><td>Moderate</td></tr>
<tr><td>27.</td><td>Create a dataframe in R for the following table
<table><tr><th>Rank</th><th>Name</th><th>Revenue</th><th>Profits</th><th>Assets</th></tr><tr><td>1</td><td>Walmart</td><td>559151</td><td>13510</td><td>252496</td></tr><tr><td>2</td><td>Amazon</td><td>386064</td><td>21331</td><td>321195</td></tr><tr><td>3</td><td>Apple</td><td>274515</td><td>57411</td><td>323888</td></tr><tr><td>4</td><td>CVS Health</td><td>268706</td><td>7179</td><td>230715</td></tr><tr><td>5</td><td>UnitedHealth Group</td><td>257141</td><td>15403</td><td>197289</td></tr></table></td><td>Moderate</td></tr>
<tr><td>28.</td><td>Following is the coding for framing the data in R. Find the 4 mistakes and rewrite the following code :
data = data-frame(Employee=c(Tara, Sita, Akash, Tarun, Gagan),
Work_exp = (10, 15, 16, 14, 16),
MobileNo=c(98000046 54002456 78940069 55987800 22564200 23569800))</td><td>Moderate</td></tr>
<tr><td>29.</td><td>Import the dataset credit_data to Rstudio and check its summary. Save it as a data frame ‘CD’. Identify the Columns having missing values.</td><td>Easy</td></tr>
<tr><td>30.</td><td>Find minimum, maximum and average price of houses from data set Indian Houses.</td><td>Easy</td></tr>
<tr><td>31.</td><td>You want to view the minimum, mean, median, and maximum values for all columns in the airquality dataset, including how many missing values are present. Apply single R command to view this?</td><td>Easy</td></tr>
<tr><td>32.</td><td>Write a command to display all unique plant names from the Plant column in the CO2 dataset.</td><td>Easy</td></tr>
</table>

Unit II: Data Wrangling

Sr. No.	Question	Difficulty level
33.	Define Data wrangling. Explain the processes in data wrangling in detail. Also, write advantages of data wrangling.	Moderate
34.	Which commands are used in the 'dplyr' package? Describe any two with their purposes and syntax with an example.	Moderate
35.	Which are commands used in package tidyr to make data tidy? Explain its uses and syntax	Moderate
36.	What is tidy data? Differentiate between purposes of gather and unite command with their syntax.	Moderate
37.	What is main objective of package 'dplyr' in data analysis process. Give any two commands from 'dplyr' with examples	Moderate
38.	Write uses and difference between commands: gather and unite.	Moderate
39.	What different commands can be used to identify the 'NA' values from the dataset? Write any two commands with proper syntax.	Easy
40.	Describe the function and syntax of the commands: 1. Slice 2. mutate 3.group_by 4.arrange 5. Summarise	Easy
Coding based questions		
For question 42-45 use "IndianHouses" data set. Write the code for following queries:		
41.	Check whether NA values are there in 'Parking' variable. Also find count of NA values.	Easy
42.	How many semi-furnished houses are having more than 4 parking? Write the command for this.	Moderate
43.	Write the code to find the count of houses as per locality?	Moderate
44.	What is average price of the Indian houses according to Transaction?	Moderate
For question 46-49 use "students_placement_Data" set. Write the code for following queries:		
45.	Convert Placement data into data frame, and verify whether minimum and maximum values of all percentage are same or different	Easy
46.	Calculate the average percentage of Operating Systems, Programming, and Algorithms observed for each suggested job role	Moderate
47.	Create a new data with following properties, Remove the unnecessary variable 32 and 36 and extract students with good coding skills rating greater than 5	Moderate
48.	Create a data without applying any filter consider observation 101 to 200 containing only percentage with suggested job	Moderate

For question 48-50 use "Placement_Data_Full_Class" data set. Write the code for following queries:

49.	What is the average salary as per the gender	Easy
50.	How many students are there who are having more than 70% in MBA; what is average salary of these students.	Easy
51.	Write the command to get count of students who are placed and completed their MBA in marketing and HR [variable is 'specialization']	Moderate
52.	What is the command to extract data from the top 30 rows of 'airquality'?	Easy
53.	Write a command to get the month wise average of wind speed for 'airquality'	Easy
54.	Refer the dataset 'healthcare_data' and write the commands to get the new data for the following: 1. Create a subset of above dataset for 'age' greater than 65 and save as H1. 2. Use H1, to find mean of 'bmi' as per categories under 'gender'.	DIFFICULT
55.	For the dataset 'airquality'-Write command to get a new data 1. Find average of 'Ozone' for wind speed greater than 7.5 2. Sort the data for "month" in descending order and remove if any NA values.	DIFFICULT
56.	Write a command to extract data , row wise from 101 to 250 rows from dataset 'mpg'	Easy
57.	If the given data is 'table1' write syntax to spread population year wise country | year | cases | population Afghanistan | 1999 | 745 | 19987071 Afghanistan | 2000 | 2666 | 20595360 Brazil | 1999 | 37737 | 172006362 Brazil | 2000 | 80488 | 174504898 China | 1999 | 212258 | 1272915272 China | 2000 | 213766 | 1280428583	Easy
58.	Refer the dataset 'diamonds' a. Filter data to get 'Premium' cut of diamonds and store to D1 b. Arrange the D1 in ascending order of 'price' and name it 'D2'	Moderate
59.	Write the code for the following • Import the given dataset of car_price_prediction, do we need to change the data type for a few variables? If yes, change the data type (Paste the code generated in console) • Extract the number of observations from 101-250 from the given dataset and create a new file 'Data1' • Generate multiple columns using Gear.box.type as a Key column and Priceas a Value column and save it as a new data frame 'Data2'.	DIFFICULT
60.	Write syntax for the following [Refer car_price_prediction] • Find Average of Mileage of as Fuel_type of car • Extract the data of 'Petrol' Fuel type having Engine volume more than 1.8 • Sort the data with respect to Prod. year as per Fuel type	Moderate
61.	Refer the dataset 'healthcare_data' and write the commands to get the new data for the following:	Moderate

- Create a subset of above dataset for ‘age’ greater than 65 and save as H1.
- Use H1, to find mean of ‘bmi’ as per categories under ‘gender’.

62. Consider the data of A data frame with 234 rows and 11 variables. It contains only models which had a new release every year between 1999 and 2008 - this was used as a proxy for the popularity of the car.

DIFFICULT

manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact
audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact
audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact
audi	a4	2.0	2008	4	auto(av)	f	21	30	p	compact

Explain the method stepwise to get a data by the manufacturer ‘audi’ and class ‘compact’ , exclude column ‘year’ and ‘drv’

63. Cluster the columns of inbuilt dataset 'billboard' into Week and its Value; where billboard dataset has 317 observations and 79 columns about Song rankings for billboard top 100 in the year 2000 as folEasys:

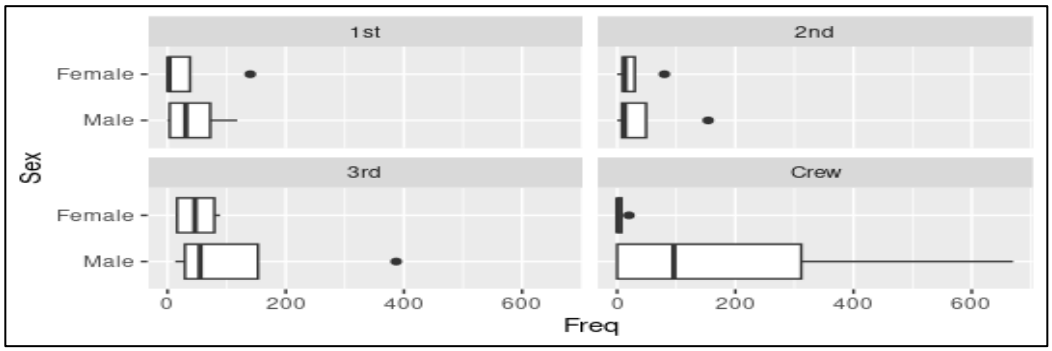
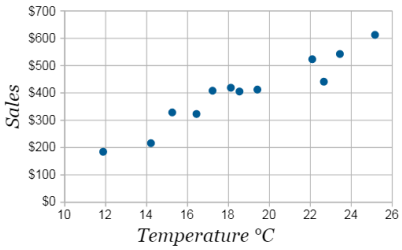
artist	track	date.entered	wk1	wk2	wk3	wk4	wk5	...	wk76
2 Pac	Baby Don't Cry (Keep..	26-02-2000	87	82	72	77	87	...	NA
2Ge+her	The Hardest Part Of ...	02-09-2000	91	87	92	NA	NA	...	NA
3 Doors Down	Kryptonite	08-04-2000	81	70	68	67	66	...	NA
3 Doors Down	Loser	21-10-2000	76	76	72	69	67	...	NA
504 Boyz	Wobble Wobble	15-04-2000	57	34	25	17	17	...	NA

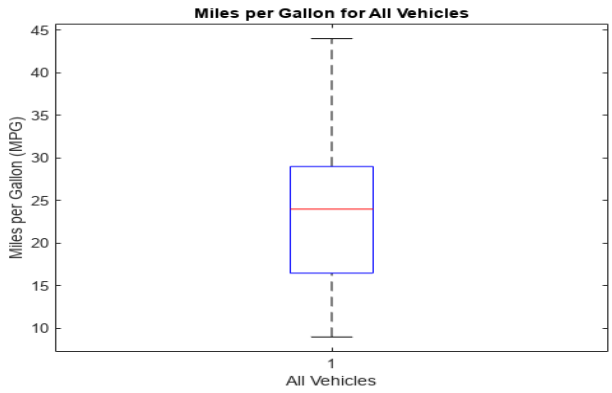
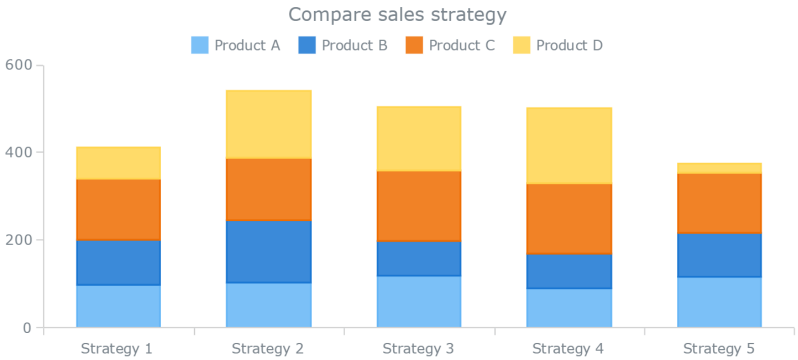
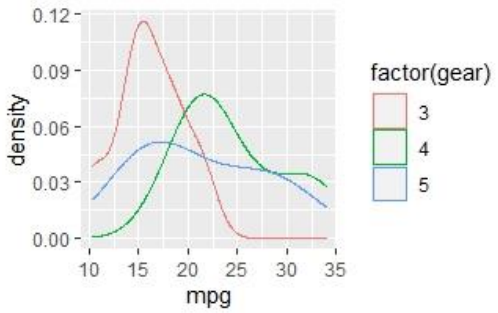
64. Following is the data for ‘IndianHouses’. Which command can be used to extract average price of houses and standard deviation as per the ‘Furnishing’ status. Write the syntax for it.

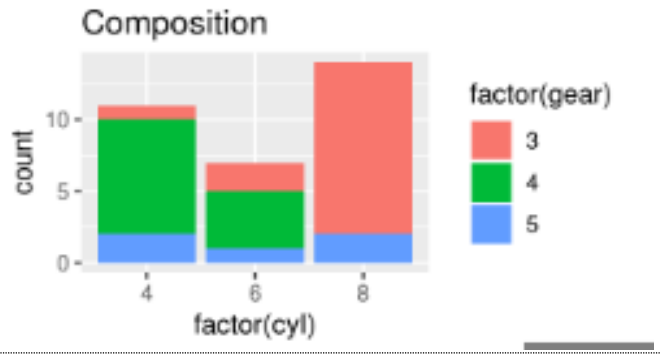
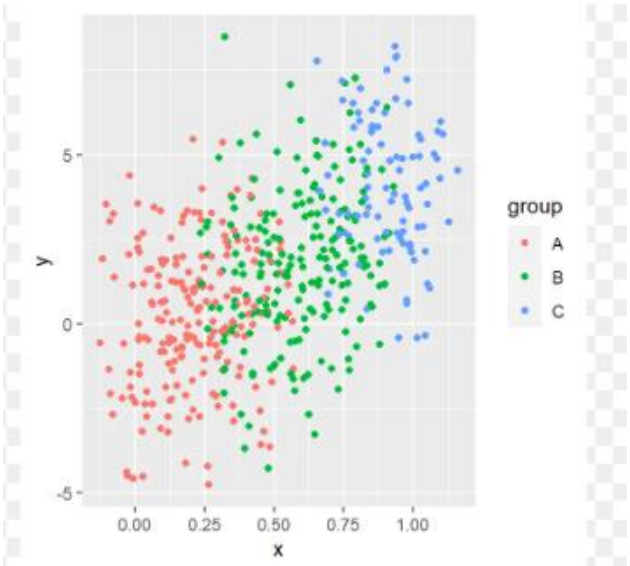
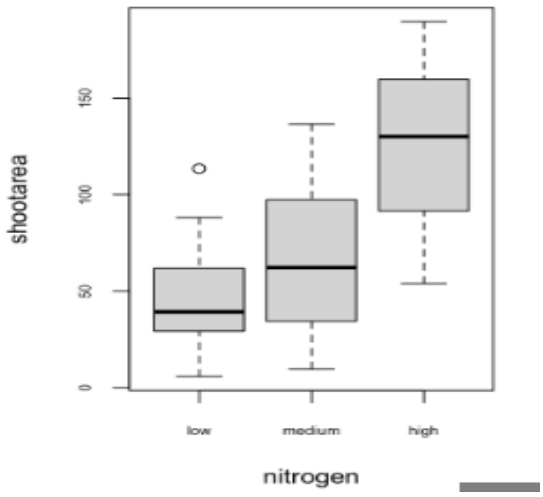
Area	BHK	Bathroom	Furnishing	Locality	Parking	Price	Status	Transaction	Type	Per_Sqft
4050	6	3	Furnished	Budh Vihar Phase 1	1	5100000	Ready_to_move	Resale	Builder_Floor	1259
900	2	2	Furnished	Baljit Nagar, Patel Nagar	1	1500000	Ready_to_move	Resale	Builder_Floor	1667
405	2	1	Semi-Furnished	Patel Nagar West	1	3950000	Ready_to_move	Resale	Apartment	1667
850	2	1	Semi-Furnished	Siddhartha Heights, Dwarka Sector 24	1	1900000	Ready_to_move	New_Property	Apartment	2235
2400	4	4	Semi-Furnished	Find Homes, Greater Kailash 2	1	7E+07	Ready_to_move	New_Property	Apartment	2235

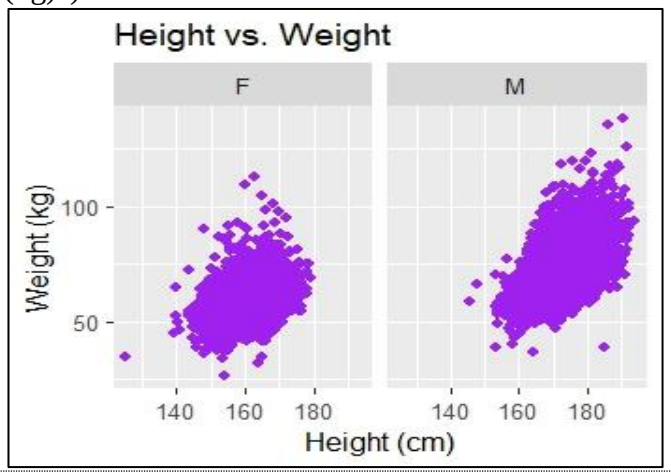
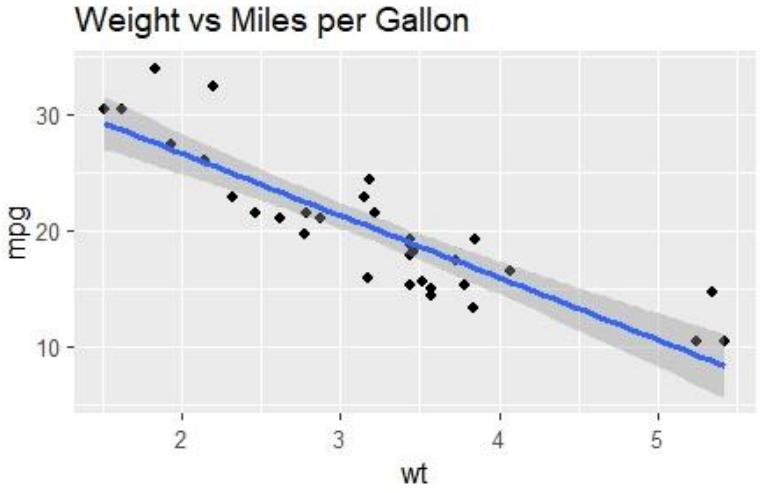
Unit III: Data Visualization in R Software

Sr. No.	Question	Difficulty level
For question 66-68 use "IndianHouses" data set. Write the code for the following		
65.	Using visualization show number of houses according to their 'Type'.	Easy
66.	Visualize the relation between area and price according to their furnishing status with title 'Relation between Area and Price'. Also comment on Area depends on price.	Easy
67.	Comment on frequency according to status of the houses and show it with graph. Also give title 'Count of House as per Status' for the graph	Moderate
For question 69-71 use "students_placement_Data" set. Write the code for the following		
68.	Find which jobs profiles are suggested more and also compare it with visualization with appropriate plot	Easy
69.	Whether suggested job roles are affected by extra courses done, visualize it and comment on the same	Moderate
70.	By visualization check whether percentage of Operating system and communication are related for this new data	Easy
For question 72-74 use "Placement_Data_Full_Class" data set. Write the code for the following		
71.	Plot the graph for 'mba_p' and 'salary' as per the 'workex' by omitting NA values if any and put suitable labels	Easy
72.	Find the trend between two variables 'etest_p' and 'mba_p', and put suitable labels. What inference can be derived from it.	Moderate
73.	Find the graph of 'specialization' as per the placement 'status' and put suitable labels	Easy

74.	Write 5 inferences from following Boxplot for Frequencies of passenger as per their gender and Class in Titanic. 	DIFFICULT
75.	Explain the following any two plots by their uses and interpretation **I.** Bar- plot **II.** Violin plot **III.** Scatter Plot	Easy
76.	Create a data frame 'Student_count' as beEasy, and get a Pie chart by ggplot showing strength of the students as per group. **group** **Num_student** Arts 30 Commerce 23 Science 16 Engineering 21	Easy
77.	500 students are enrolled for sports as folEasys. Write the code to describe the participation with Pie chart. Also, tell significance of Pie chart for Data visualization. **Sports** / **No. of students participating** Cricket 121 Baseball 90 Hockey 110 Atheletics 91 Chess 88	Moderate
78.	What is Jitter plot? Explain uses of it? Write a syntax to get following Jitter plot for variable 'cyl' and 'displ' of 'mpg' data. If the same command run multiple time is there any change observed in output plot?	Moderate
79.	Write interpretation of the following plot 	DIFFICULT
80.	Write interpretation of the following plot	DIFFICULT

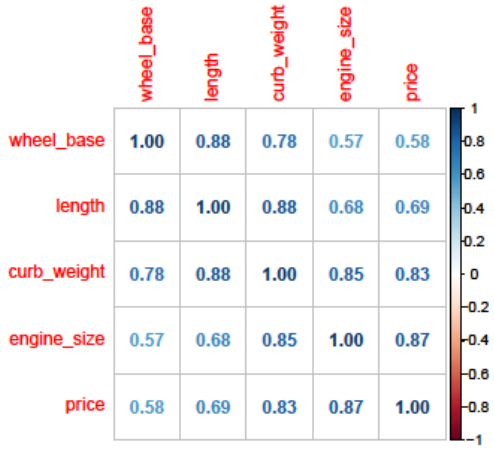
		
81.	Write interpretation of the following plot 	DIFFICULT
82.	Write interpretation of the following plot 	DIFFICULT
83.	Differentiate between the following plots as per the data type also explain how this plot can be interpreted 1. Boxplot 2. Histogramme 3. Barplot 4. Scatterplot	Easy
84.	Why ggplot is used as an effective package for data visulization? Describe all the components of ggplot	Moderate
85.	Write syntax to get the following plot for data ‘mpg’	Easy

		
86.	Write syntax to get the following plot for data 'Sales' 	Moderate
87.	Compare boxplot in the given plot 	DIFFICULT
88.	Density Plot showing 'price' distribution as per 'number_of_doors' and coefficient of skewness for price distribution as per number of doors is given for the data set 'Car_Price'. Write a syntax to get following plots and coefficient of skewness for price as per number of doors and also comment on the skewness of the price **num_of_doors** | **Coeff_Skewness** four | 1.574059 two | 1.935925	DIFFICULT

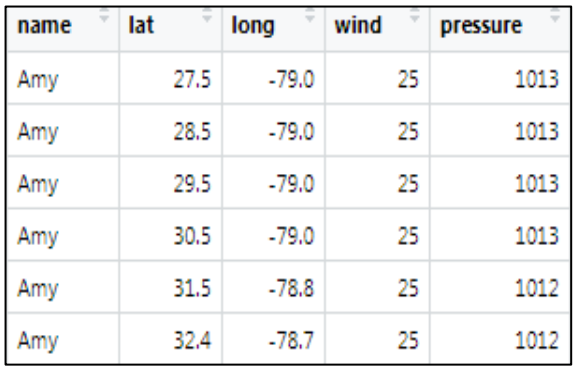
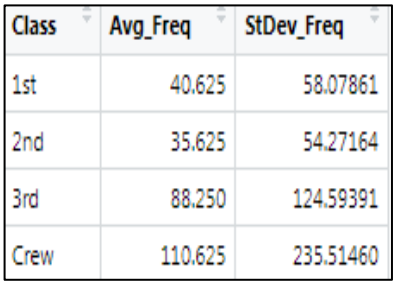
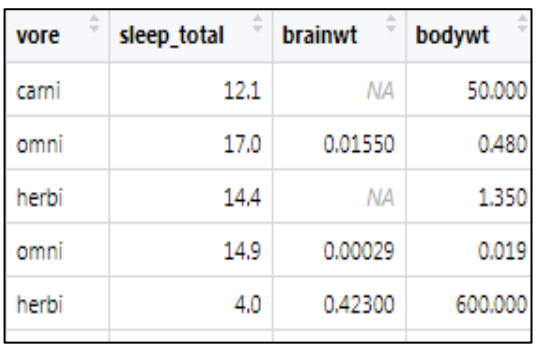
89.	Find the mistake in the code and rewrite again to generate the given graph. Also write inferences for it. ggplot(bodyPerformance, aes(x = height_cm, y = weight_kg)) + geom(color = "purple") +facet_wrap(gender)+ labs(title = Height vs. Weight, xlab = "Height (cm)", ylab = "Weight (kg)") 	DIFFICULT
90.	Write the inferences for the following plot for 'mtcars' dataset where mpg is miles per gallon and wt is weight. Also, write the command to obtain the given graph 	DIFFICULT
Use "Diamond" data for question no. 92-103		
91.	Create a data to get count of diamonds to get color variations across different clarity grades.	Easy
92.	How many diamonds exist in the dataset for each color grade, compare it.	Easy
93.	How does the distribution of diamond color vary across different clarity grades, and which color grades are more in all clarity categories?	Moderate
94.	What is the relationship between a diamond's carat weight and its price? Does the price increase linearly with carat?	Moderate
95.	What is the relationship between a diamond's carat weight and its price? Does the price increase linearly with carat?	Moderate
96.	Does the impact of carat weight on price differ depending on the diamond's cut quality (Fair, Good, Very Good, etc.)?	Easy

97.	Visualize How does the quality of a diamond's cut (Fair, Good, Very Good, Premium, and Ideal) affect its price?	Moderate
98.	Is there a significant difference in price between diamonds with different clarity grades?	Moderate
99.	How does a diamond's color affect its price? Are certain color grades linked to much higher prices compared to others?	Moderate
100.	What combination of carat, cut, color, and clarity results in the highest diamond prices?	Easy
101.	How does the interaction between clarity and cut quality affect diamond prices?	Moderate
102.	Is there any relationship between a diamond's carat and its clarity grade? Are larger diamonds more likely to have Easyer clarity?	Moderate

Unit IV: Data Analysis

Sr. No.	Question	Difficulty level
103.	What is the role of measures of central tendency in descriptive statistics? What are the criteria to use either of mean, median or mode as central tendency for the given data?	Moderate
104.	What is significance of measure of variability in Data analysis. Explain standard deviation in detail.	Moderate
105.	What is significance of skewness and kurtosis in data analysis	Easy
106.	For data set 'Car_price' write a syntax to get correlation of the following variable and also beEasy correlation plot. Comment on the correlation of all variables with 'price' 	Moderate
107.	If for two groups of employees, data of dispersion for High-Income and Easy-income Earners is as follows. If the number of observation are same in both the groups which measure of dispersion to be used. Write inferences from the following table.	Easy

Standard Deviation	100,000	5,000
Coefficient of Variation	16.66	14.28
Mean	600,000	35,000

108.	A data set provides measurements of the diameter, height and volume of timber in 31 felled black cherry trees. Following is data of descriptive analysis Girth Height Volume ---------- -------- -------- -------- Std.Dev 3.14 6.37 16.44 Min 8.30 63.00 10.20 Q1 11.00 72.00 19.10 Median 12.90 76.00 24.20 Q3 16.00 80.00 38.30 Max 20.60 87.00 77.00 MAD 2.82 5.93 11.56 IQR 4.20 8.00 17.90 CV 0.24 0.08 0.54 Make inferences from dispersion measures for variables Girth(Diameter), height and Volume	Moderate
109.	Write a program to get correlation matrix for the data 'storms' given beEasy. Assume that there is NA values present in the data. Also find corrplot of with correlation matrix. (Mention required package) 	DIFFICULT
110.	Write syntax to get average of 'Freq' as per the 'Class' for dataset 'Titanic; which has shown in table. Freq shows number of passengers as per class. Comment on variation in frequencies Class wise 	Moderate
111.	First 5 entries from data 'msleep' related to the mammals sleep dataset of carnivore, omnivore or herbivore. Write a syntax to visualize the relation between sleep_total and bodywt. Also comment on it. 	DIFFICULT

name	lat	long	wind	pressure
Amy	27.5	-79.0	25	1013
Amy	28.5	-79.0	25	1013
Amy	29.5	-79.0	25	1013
Amy	30.5	-79.0	25	1013
Amy	31.5	-78.8	25	1012
Amy	32.4	-78.7	25	1012

Class	Avg_Freq	StDev_Freq
1st	40.625	58.07861
2nd	35.625	54.27164
3rd	88.250	124.59391
Crew	110.625	235.51460

vore	sleep_total	brainwt	bodywt
carni	12.1	NA	50.000
omni	17.0	0.01550	0.480
herbi	14.4	NA	1.350
omni	14.9	0.00029	0.019
herbi	4.0	0.42300	600.000

112. Write a syntax to get average of 'sleep_total' and average & coefficient of variation of 'bodywt' as per eating nature(vore). Compare variation of bodywt as per vore type. — DIFFICULT

vore	Count	Average_sleep	Avg_weight	CV_bodywt
carni	19	10.378947	90.75111	2.006266
herbi	32	9.509375	366.87725	3.390994
insecti	5	14.940000	12.92160	2.042203
omni	20	10.925000	12.71800	1.941643
NA	7	10.185714	0.85800	1.560307

113. How Skewness and Kurtosis will help to describe distribution of the data.
Write syntax to get values of Skewness and Kurtosis for distribution of variable 'wind' [that describes speed of wind] of the dataset 'storms'. — Moderate

114. If for two groups of employees, data of dispersion for High-Income and Easy-income Earners is as followss. If the number of observation are same in both the groups which measure of dispersion to be used. Write inferences from the following table. — Easy

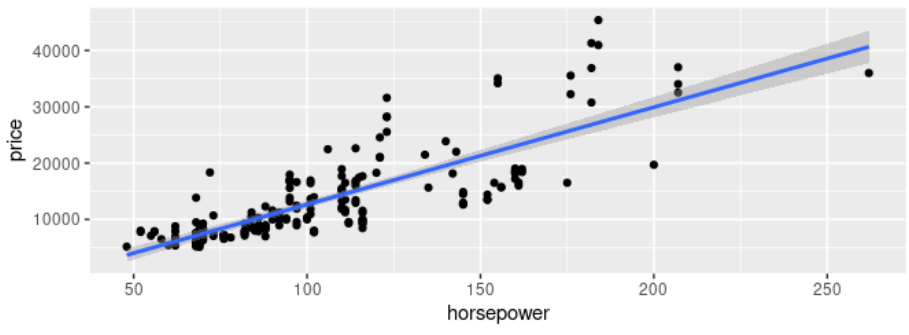
	High-Income Earners	**Easy-income Earners**
Standard Deviation	100,000	5,000
Coefficient of Variation	16.66	14.28
Mean	600,000	35,000

115. Correlation matrix of 'storms' data is given beEasy — Moderate

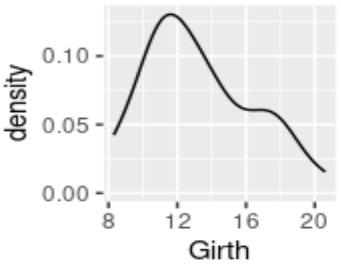
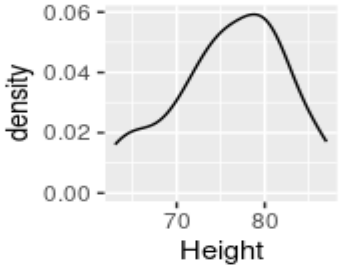
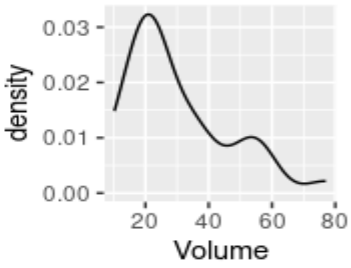
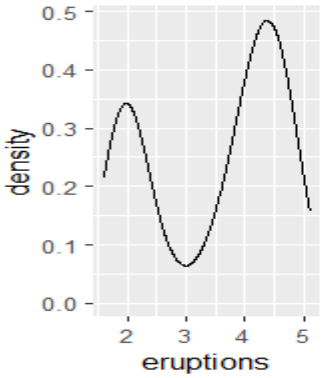
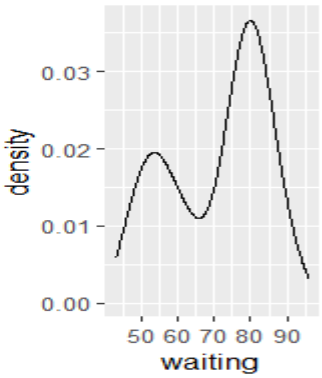
```
                wind    pressure         lat        long
wind       1.000000000 -0.92918148 -0.007004468 -0.03231183
pressure  -0.929181478  1.00000000 -0.109845647  0.04086079
lat       -0.007004468 -0.10984565  1.000000000  0.11516677
long      -0.032311827  0.04086079  0.115166770  1.00000000
```

Write inferences from the correlation matrix. (where lat- latitude and long- longitude denotes location of storms

116. Describe the relationship between the variables 'car_price' and 'horsepower' of engine in the following scatter plot, also write syntax to get following plot if data set name is 'Car_price' — Easy



117. Explain uses of qnorm, pnorm and dnorm commands for probability distribution. — Moderate

118.	A data set provides measurements of the diameter, height and volume of timber in 31 felled black cherry trees, coefficient of kurtosis and density curves are given beEasy    Girth Height Volume Kurtosis -0.71 -0.72 0.25 Write minimum 3 comments on distribution of the data based on kurtosis	DIFFICULT
119.	'faithful' is the data set involving 'waiting' [time between eruptions] and 'eruptions' [the duration of the eruptions] for the Old Faithful geyser. Statistical Analysis is given in the table as followss. Also, related graphs are given. Write the inferences based on skewness and kurtosis. **Skewness** **Kurtosis** **Mean** **Std. Deviation** **eruptions** -0.4158 1.4994 3.4880 1.1413 **waiting** -0.4163 1.8574 70.9000 13.5949  	DIFFICULT
120.	The CO2 uptake and concentration of six plants from 'Quebec' and six plants from 'Mississippi' was measured at several levels of ambient CO2. Correlation matrix plantwise is given beEasy. Write a syntax to get this output and also comment on correlation. Correlation for Quebec plants Conc Uptake Conc 1. 000 0.710 uptake 0.710 1.000 Correlation for Mississippi plants Conc Uptake Conc 1. 000 0.471 uptake 0.471 1.00	DIFFICULT
121.	Write a syntax to get following statistical analysis of average of 'sleep_total' and 'bodywt',similarly coefficient of variation ' bodywt' of as per eating nature(vore) of the 'msleep' dataset. Compare variation of bodywt as per vore type. **vore** **Avg.sleep** **Avg.bodywt** **COV.bodywt** carni 10.3789 90.7511 200.6266 herbi 9.5094 366.8773 339.0994 insecti 14.9400 12.9216 204.2203 omni 10.9250 12.7180 194.1643 NA 10.1857 0.8580 156.0307	Difficult

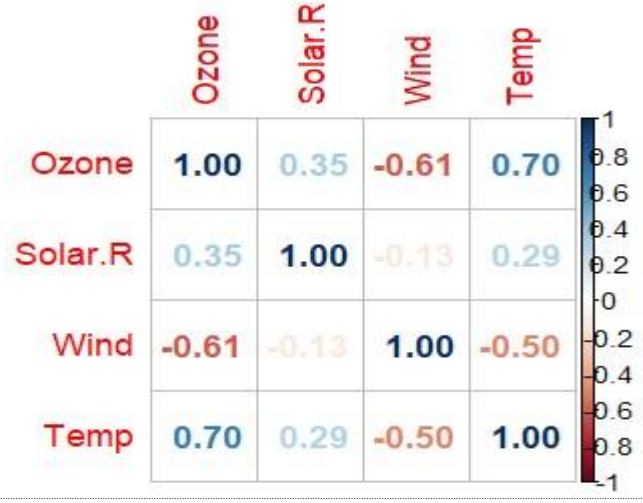
122.	Correlation matrix of 'storms data' is given beEasy
	`wind pressure lat long`
	`wind 1.000000000 -0.92918148 -0.007004468 -0.03231183`
	`pressure -0.929181478 1.00000000 -0.109845647 0.04086079`
	`lat -0.007004468 -0.10984565 1.000000000 0.11516677`
	`long -0.032311827 0.04086079 0.115166770 1.00000000`
	Write inferences from the correlation matrix. (where lat- latitude and long- longitude denotes location of storms
		Moderate

123.	What is the difference between standard deviation and coefficient of variation? After conducting a systematic investigation on high-income earners and Easy-income earners in a community, a researcher has the following results: **High-Income Earners** / **Easy-income Earners** Standard Deviation: $100,000 / $5,000 Coefficient of Variation: 16.66 / 14.28 Mean: $600,000 / $35,000 Then Analyse the outcomes.	DIFFICULT
124.	A data set provides measurements of the diameter, height and volume of timber in 31 felled black cherry trees. Standard deviation of diameter, height and volume is as given beEasy **Sd_Diam** **Sd_Height** **Sd_Volume** 3.138139 6.371813 16.43785 Also its graphical representation in histogram is given beEasy Comment about the central tendency and dispersion in data of all three variables	Moderate
125.	What is the correlation coefficient? How would it help in the understanding of the relationship between two variables?	Easy

126.	A data set provides measurements of the diameter, height and volume of timber in 31 felled black cherry trees. Following is Correlation Matrix for numerical data
	`Diameter Height Volume`
	`Diameter 1.0000000 0.5192801 0.9671194`
	`Height 0.5192801 1.0000000 0.5982497`
	`Volume 0.9671194 0.5982497 1.0000000`
	Comment about correlation ship among variables
		Moderate

127.	Take into account the data beEasy (As a sample first 6 observations are given), where there are **5** different types of trees
	`Tree age circumference`
	`1 1 118 30`
	`2 1 484 58`
	`3 1 664 87`
	`4 1 1004 115`
	`5 1 1231 120`
	`6 1 1372 142`
	Create a programme that will yield the mean, median, and standard deviation for each form of tree.
		DIFFICULT

128.	Write the code to generate the following corrplot for 'mtcars' data set, where mpg: miles per gallon, cyl: number of cylinders, hp: horsepower of engine, wt:weight of engine. Comment on the correlation of mpg with other variables.	Moderate

		
129.	Write the code to generate the following corrplot for 'airquality' data set. Comment on the correlation of Ozone with other variables. 	Easy

Sr. No.	Question	Difficulty level
130.	What are the assumptions to apply a regression model? What is impact on regression model if assumptions are failed.	Easy
131.	What is regression analysis? What is importance of regression analysis in data analysis. Write syntax to find regression analysis between one dependent and one independent variable from a dataset	Moderate
132.	What is use of sampling in inferential statistics? Explain methods of sampling.	Easy
133.	What is difference between probability and nonprobability sampling?	Easy

134.	A regression model is applied on 'storms' data set for variable 'Pressure' as dependant variable and 'wind' as independent variable.
	Write syntax to get this regression model
	Write Null hypothesis and inferences from the given summary of the regression model:
	Regression model:
	`Coefficients:`
	`Estimate Std. Error t value Pr(>|t|)`
	`(Intercept) 1.028e+03 1.104e-01 9306.2 <2e-16 ***`
	`wind -6.827e-01 1.967e-03 -347.1 <2e-16 ***`
	`---`
	`Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1`
	`Residual standard error: 6.926 on 19064 degrees of freedom`
	`Multiple R-squared: 0.8634, Adjusted R-squared: 0.8634`
	`F-statistic: 1.205e+05 on 1 and 19064 DF, p-value: < 2.2e-16`
		Difficult

135.	For variables concentration and uptake a data set 'CO2', summary of the regression model is given below. Comment on dispersion of residual, accuracy of the model and dependency of the variable *Residuals:* *Min 1Q Median 3Q Max* *-358.75 -177.93 -92.14 140.33 735.11* *Residual standard error: 260.3 on 82 degrees of freedom* *Multiple R-squared: 0.2354, Adjusted R-squared: 0.2261* *F-statistic: 25.25 on 1 and 82 DF, p-value: 2.906e-06*	Difficult

136.	Comment on beEasy outputs of simple linear regression model for '*p value'* , '*R Squared*', residual and their distribution
	`Call:`
	`lm(formula = Volume ~ Height, data = trees)`
	`Residuals:`
	`Min 1Q Median 3Q Max`
	`-21.274 -9.894 -2.894 12.068 29.852`
	`Coefficients:`
	`Estimate Std. Error t value Pr(>|t|)`
	`(Intercept) -87.1236 29.2731 -2.976 0.005835 **`
	`Height 1.5433 0.3839 4.021 0.000378 ***`
	`---`
	`Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1`
	`Residual standard error: 13.4 on 29 degrees of freedom`
	`Multiple R-squared: 0.3579, Adjusted R-squared: 0.3358`
	`F-statistic: 16.16 on 1 and 29 DF, p-value: 0.0003784`
		Difficult

137.	Following is the output of regression model applied for 'airquality' dataset where Ozone is dependent variable and Temp and Wind are independent variables. Write syntax for this regression model and also write the inferences from the given summ ary
	Residuals:
	`Min 1Q Median 3Q Max`
	`-41.251 -13.695 -2.856 11.390 100.367`
	Coefficients:
	`Estimate Std. Error t value Pr(>|t|)`
	`(Intercept) -71.0332 23.5780 -3.013 0.0032 **`
	`Temp 1.8402 0.2500 7.362 3.15e-11 ***`
	`Wind -3.0555 0.6633 -4.607 1.08e-05 ***`
	`---`
	Signif. codes: `0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1`
		Difficult

```
Residual standard error: 21.85 on 113 degrees of freedom  (3
7 observations deleted due to missingness)
Multiple R-squared:  0.5687,       Adjusted R-squared:  0.561
1
F-statistic:  74.5 on 2 and 113 DF,  p-value: < 2.2e-16
```